

7.2 PRIME FACTORIZATION



prime number - a number _____ than _____ and has _____ two factors
1 and itself

Prime numbers are the _____ of all numbers.

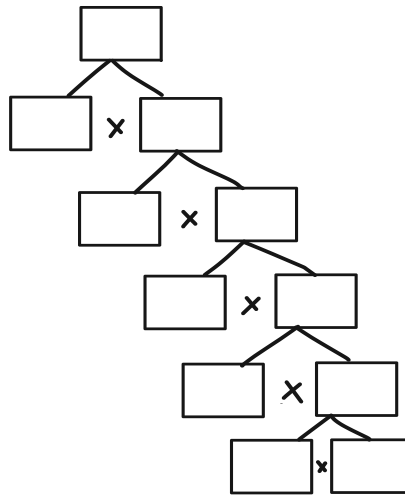
FIND the PRIME FACTORIZATION of 600

Two WAYS to SHOW prime factorization

1. Ladder Diagram



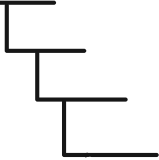
2. Tree Diagram



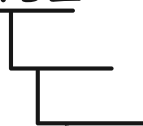
Prime Factorization of 600 →

Find the Prime Factorization of these numbers.

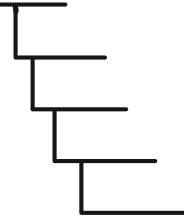
32



132



144



7.2 PRIME FACTORIZATION

FACTOR TREES

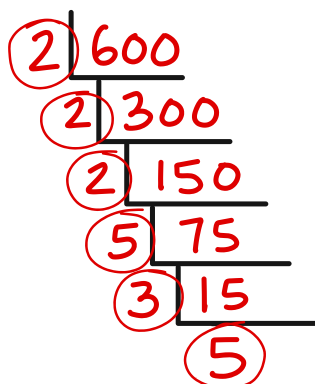
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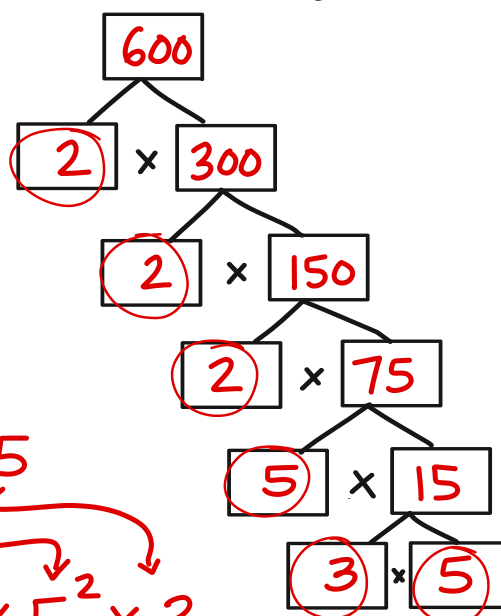
Two WAYS to SHOW prime factorization

1. Ladder Diagram



means
divide

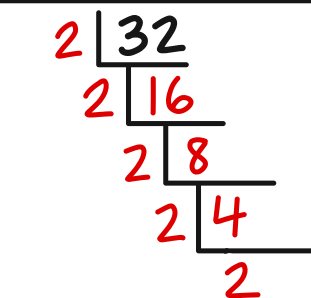
2. Tree Diagram



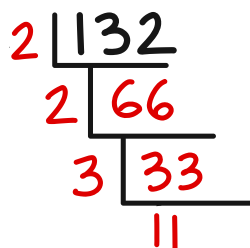
$$2 \times 2 \times 2 \times 5 \times 3 \times 5$$

Prime Factorization of 600 $\rightarrow 2^3 \times 5^2 \times 3$

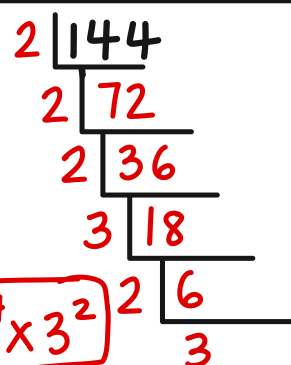
Find the Prime Factorization of these numbers.



$$2 \times 2 \times 2 \times 2 \times 2 = 2^5$$



$$2 \times 2 \times 3 \times 11 = 2^2 \times 3 \times 11$$



$$2^4 \times 3^2$$

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FACTOR TREES

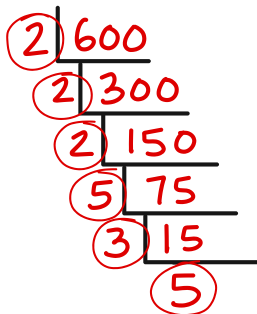
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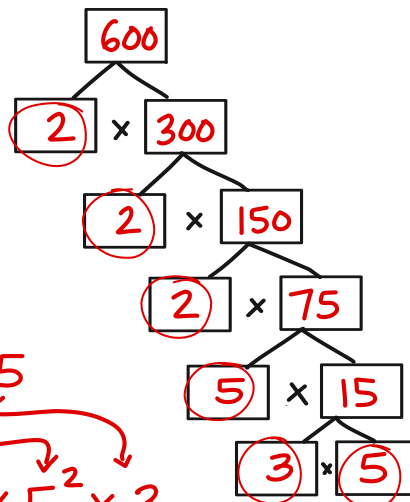
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2. Tree Diagram

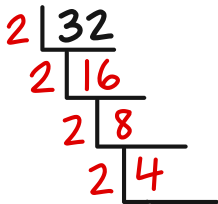


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divide

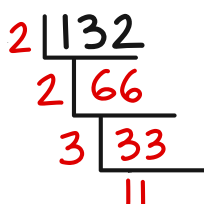
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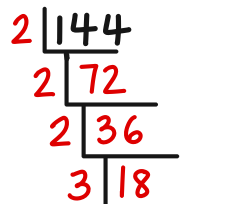
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cut on dashed lines and glue in notebook

7.2 PRIME FACTORIZATION



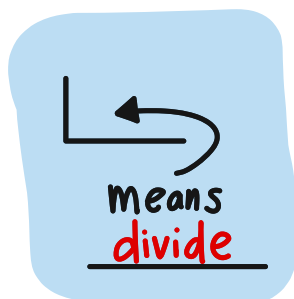
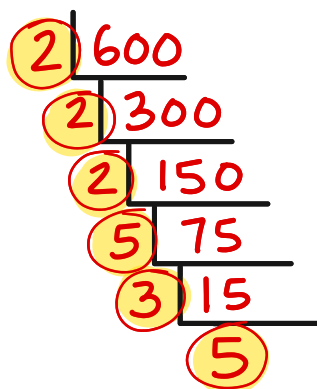
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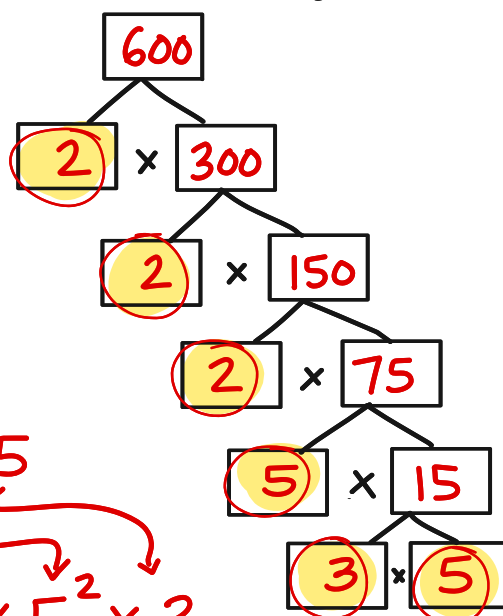
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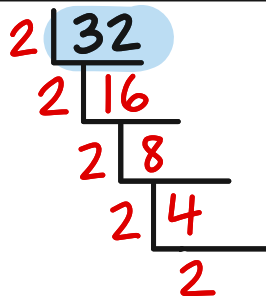
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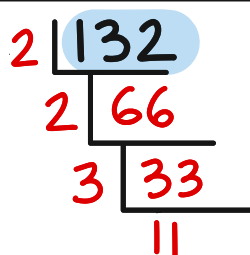
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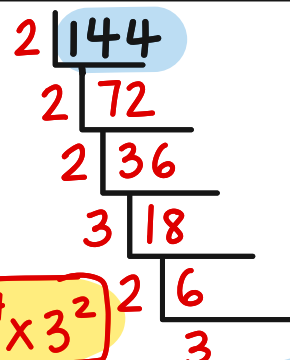
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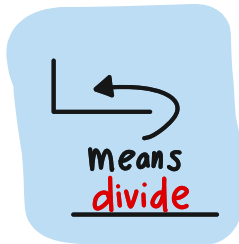
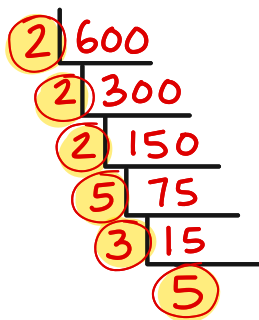
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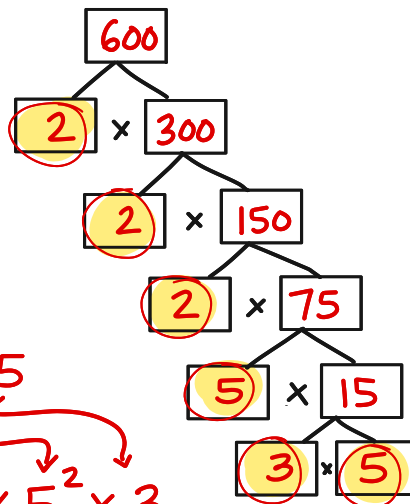
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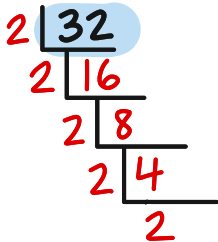
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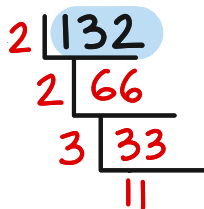
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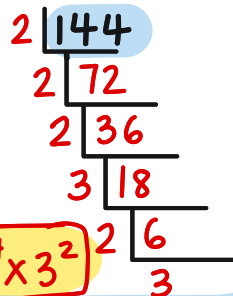
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